

Activity 1:

Sample	Mean	Sample	Mean	Sample	Mean
4, 4, 4	4	6, 4, 4	4.67	5, 4, 4	4.33
4, 4, 6	4.67	6, 4, 6	5.33	5, 4, 6	5
4, 4, 5	4.33	6, 4, 5	5	5, 4, 5	4.67
4, 6, 4	4.67	6, 6, 4	5.33	5, 6, 4	5
4, 6, 6	5.33	6, 6, 6	6	5, 6, 6	5.67
4, 6, 5	5	6, 6, 5	5.67	5, 6, 5	5.33
4, 5, 4	4.33	6, 5, 4	5	5, 5, 4	4.67
4, 5, 6	5	6, 5, 6	5.67	5, 5, 6	5.33
4, 5, 5	4.67	6, 5, 5	5.33	5, 5, 5	5

X	4.00	4.33	4.67	5	5.33	5.67	6
$P(X)$	1.) $1/27 = 0.04$	2.) $3/27 = 0.11$	3.) $6/27 = 0.22$	4.) $7/27 = 0.26$	5.) $6/27 = 0.22$	6.) $3/27 = 0.11$	7.) $1/27 = 0.04$

8.) Mean = $\Sigma(X \times P(X))$
= $(4.00 \times 0.04) + (4.33 \times 0.11) + (4.67 \times 0.22) + (5.00 \times 0.26) + (5.33 \times 0.22) + (5.67 \times 0.11) + (6.00 \times 0.04) = 5$

9.) Variance = $\Sigma((X - \text{Mean})^2 \times P(X))$
 $00 - 5.00)^2 \times 0.04) + ((4.33 - 5.00)^2 \times 0.11) + ((4.67 - 5.00)^2 \times 0.22) + ((5.00 - 5.00)^2 \times 0.26) + ((5.33 - 5.00)^2 \times 0.22) + ((5.67 - 5.00)^2 \times 0.11) + ((6.00 - 5.00)^2 \times 0.04)$
= 0.23

10.) Standard deviation = $\sqrt{\text{Variance}}$
 $\sqrt{0.23} = 0.48$

Activity 2:

- 1.) $\mu - \mu\bar{x} = 5 - \underline{5} = \underline{0}$
- 2.) $\sigma - \sigma\bar{x} = 0.82 - \underline{0.48} = \underline{0.34}$
- 3.) Population standard deviation > Standard deviation of the sample means
- 4.) $0.82/1.73 = 0.474$ or 0.48
- 5.) Yes

Activity 3:

Given data: {35, 40, 37, 38, 40}

1. Mean (μ):

$$\text{Mean} = \frac{\sum_{i=1}^n X_i}{n}$$

$$\text{Mean} = \frac{35+40+37+38+40}{5}$$

$$\text{Mean} = \frac{190}{5}$$

$$\text{Mean} = 38$$

2. Variance (σ^2):

$$\text{Variance} = \frac{\sum_{i=1}^n (X_i - \text{Mean})^2}{n}$$

$$\text{Variance} = \frac{(35-38)^2 + (40-38)^2 + (37-38)^2 + (38-38)^2 + (40-38)^2}{5}$$

$$\text{Variance} = \frac{9+4+1+0+4}{5}$$

$$\text{Variance} = \frac{18}{5}$$

$$\text{Variance} = 3.6$$

3. Standard Deviation (σ):

$$\text{Standard Deviation} = \sqrt{\text{Variance}}$$

$$\text{Standard Deviation} = \sqrt{3.6}$$

$$\text{Standard Deviation} \approx 1.897$$

4-8.)

Sample	x	Sample	x	Sample	x	Sample	x	Sample	x
35, 35	35	40, 35	37.5	37, 35	36	38, 35	36.5	40, 35	37.5
35, 40	37.5	40, 40	40	37, 40	38.5	38, 40	39	40, 40	40
35, 37	36	40, 37	38.5	37, 37	37	38, 37	37.5	40, 37	38.5
35, 38	36.5	40, 38	39	37, 38	37.5	38, 38	38	40, 38	39
35, 40	37.5	40, 40	40	37, 40	38.5	38, 40	39	40, 40	40

9-12.)

35	35.5	36	36.5	37	37.5	38	38.5	39	39.5	40
1/25 or 0.04	0	2/25 or 0.08	2/25 or 0.08	1/25 or 0.04	6/25 or 0.24	1/25 or 0.04	4/25 or 0.16	4/25 or 0.16	0	4/25 or 0.16

- 13.)
- Mean = $\Sigma(X * P(X))$
= $(35 * 0.04) + (36 * 0.08) + (36.5 * 0.08) + (37 * 0.04) + (37.5 * 0.24) + (38 * 0.04) + (38.5 * 0.16) + (39 * 0.16) + (40 * 0.16)$
≈ 37.925
- 14.)
- Variance = $\Sigma((X - \text{Mean})^2 * P(X))$
= $((35 - 37.925)^2 * 0.04) + ((36 - 37.925)^2 * 0.08) + ((36.5 - 37.925)^2 * 0.08) + ((37 - 37.925)^2 * 0.04) + ((37.5 - 37.925)^2 * 0.24) + ((38 - 37.925)^2 * 0.04) + ((38.5 - 37.925)^2 * 0.16) + ((39 - 37.925)^2 * 0.16) + ((39.5 - 37.925)^2 * 0) + ((40 - 37.925)^2 * 0.16)$
≈ 1.228475
- 15.)
- Standard deviation = $\sqrt{\text{Variance}}$
≈ $\sqrt{1.228475}$
≈ 1.108

New Module

Activity 1:

- 😊 True
- 😊 True
- 😬 False
- 😊 True
- 😬 False

Activity 2:

- 1.) Given: $n = 36, \mu = 18, \sigma = 6.5$

Solution:

$$\mu_{\bar{x}} = \mu = \mu_{\bar{x}} = 18$$

$$\sigma_{\bar{x}} = 6.5 / \sqrt{36} = 6.5 / 6 \quad \sigma_{\bar{x}} = 1.08$$

- 2.) Given: $n = 49, \mu = 28.6, \sigma = 12$

Solution:

$$\mu_{\bar{x}} = \mu = \mu_{\bar{x}} = 28.6$$

$$\sigma_{\bar{x}} = 12 / \sqrt{49} = 12 / 7 \quad \sigma_{\bar{x}} = 1.71$$

- 3.) Given: $n = 36, \mu = 120, \sigma = 20$

Solution:

$$\mu_{\bar{x}} = \mu = \mu_{\bar{x}} = 120$$

$$\sigma_{\bar{x}} = 20 / \sqrt{36} = 20 / 6 \quad \sigma_{\bar{x}} = 3.33$$

- 4.) Given: $n = 100, \mu = 28.6, \sigma = 25$

Solution:

$$\mu_{\bar{x}} = \mu = \mu_{\bar{x}} = 28.6$$

$$\sigma_{\bar{x}} = 25 / \sqrt{100} = 25 / 10 \quad \sigma_{\bar{x}} = 2.5$$

Activity 3:

Given: $n = 40, \mu = 218.4, \sigma = 25$

a.) $\mu_{\bar{x}} = \mu$
 $\mu_{\bar{x}} = 218.4$

b.) $\sigma_{\bar{x}} = \sigma / \sqrt{n}$

$$\sigma_{\bar{x}} = 25/\sqrt{40}$$

$$\sigma_{\bar{x}} = 25/6.32$$

$$\sigma_{\bar{x}} = \mathbf{3.95 \text{ pounds}}$$

F	R	E	E	I	D	E	N	S	H	Z	E	M
I	D	A	Y	B	O	S	U	E	I	L	P	O
T	L	O	G	A	O	O	A	P	I	T	E	D
N	W	E	Z	Z	R	R	I	A	Y	X	R	E
E	E	S	U	O	T	N	R	T	E	K	A	R
S	X	X	G	R	I	T	I	R	E	L	I	A
S	E	I	A	S	X	S	C	R	E	M	I	T
B	V	T	B	E	N	I	G	N	I	V	I	L
U	E	T	O	E	S	H	O	U	S	E	S	L
L	O	G	T	E	R	E	G	E	O	H	E	U
Y	O	N	T	A	S	S	E	S	I	N	G	S
L	I	S	E	I	T	I	V	I	T	C	A	T